

Climate 5000 Series - Gen 4

9K BTU Single Zone Regular System

1-Way Cassette IDU

Engineering
Submittal
Sheet



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Overview and Project Information



Project name

Location

Architect

Engineer

General contractor

Reference #

9K Btu Single Zone Regular System - 1-Way Cassette IDU

Indoor part number	8-733-962-666
Indoor model number	BMS500-AAU009-1AHZXD
Outdoor part number	8-733-962-684
Outdoor model number	BMS500-AAS009-1CSXRD

Efficiency Data		AHRI 210/240 2023
SEER2		22.1
EER2		14.4
HSPF2 Region IV		11.8
HSPF2 Region V		9.0
COP(47F)		3.20
COP (5F)		2.18

Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	3700~11400
	Rated Capacity	9000
	Current	4.2
	Input Power	625
Heating	Capacity Range	4100~14300
	Rated Capacity (47F)	11000
	Rated Capacity (5F)	8000
	Current	4.4
	Input Power	1007

Electrical Data	
Power supply (V / Ø / Hz)	208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)	3.0 / 12.0
Max. fuse indoor/outdoor (A)	15.0 / 15.0

Compressor Data	
Type:	Rotary
Refrigerant oil /oil charge (ml)	VG74 / 310
Input (W)	325 / 826
Rated current (RLA/A)	2.40 / 5.65

Piping	
Refrigerant precharge (oz/kg)	32.45 / 0.92
Length for precharge (ft/m)	24.6 / 7.5
Liquid side (in/mm)	1/4" / Ø6.35
Gas side (in/mm)	3/8" / Ø9.52
Additional Charge of Refrigerant (oz/ft)/(g/m)	0.16 / 15
Max. refrigerant pipe length (ft/m)	82.00 / 25
Max. difference in height (ft/m)	49.20 / 15

Operation Range

ODU	Cooling (°F / °C)	-13~122 / -25~50
	Heating (°F / °C)	-13~75 / -25~24
IDU	Cooling (°F / °C)	60~90 / 16~32
	Heating (°F / °C)	32~86 / 0~30

System Data

Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		2.11
IDU	Fan motor model	ZKFN-30-8-43
	Fan motor qty (ea)	1
	Fan motor input (W)	46.0
	Fan speed (RPM)	1024 / 944 / 864
	Air flow, Turbo / H / M / L / Si (CFM)	341.4 / 294.3 / 259.0 / 235.4 / N/A
	Approx. Air Throw Range ft (m)	23.0 (7)
	Noise level, Turbo / H / M / L / Si (dB(A))	46.5 / 38.5 / 34.0 / 24.0 / 22.0
	Net weight (lbs / kg)	45.19 / 20.5
ODU	Gross weight (lbs / kg)	82.45 / 37.4
	Drain pipe diameter (mm)	ODΦ25mm
	Fan motor model	ZKFN-34-10-1L
	Fan motor qty (ea)	1
	Fan motor input (W)	50
	Fan motor speed (RPM)	800 / 700 / 600
	Air flow, Max (CFM)	1236.1
	Noise level, Max (dB(A))	54.5
	Net weight (lbs / kg)	62.17 / 28.2
	Gross weight (lbs / kg)	67.68 / 30.7

Accessories

BMS-WT2-XXD (8733962707)	Wired Wall Thermostat
BMS-DRY-XXD (8733969263)	24V Interface
BMS-PK1-7X1 (8733951010)	25ft Lineset

Standard Parts

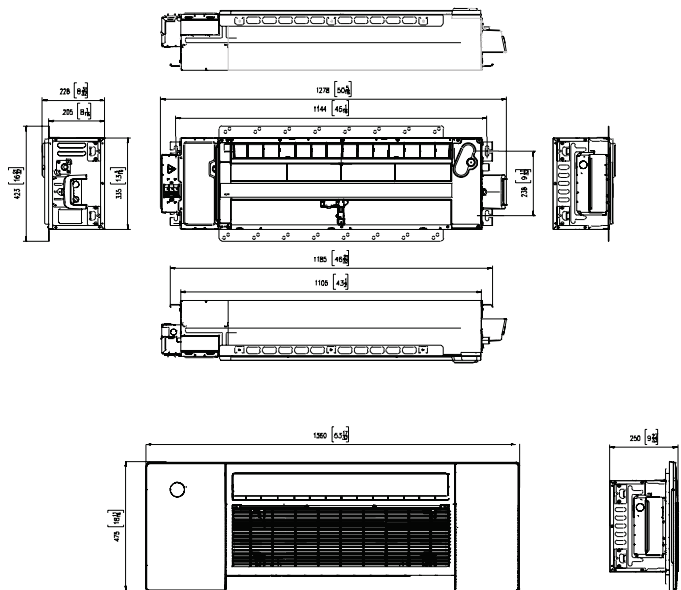
8-733-969-065	Remote Control
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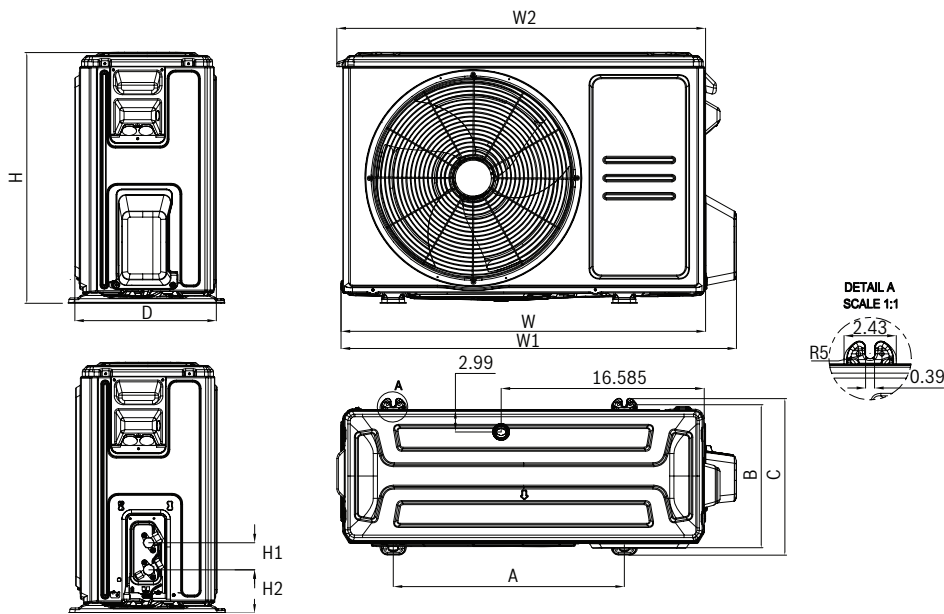


Indoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)
BMS500-AAU009-1AHZXD	50.31	13.19	8.98

Outdoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	A (in)	B (in)	C (in)	H1 (in)	H2 (in)
BMS500-AAS009-1CSXRD	31-3/4	13	21.85	34-3/8	32-1/8	20-1/8	12-1/2	13-5/8	2-3/8	3-3/4

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Cooling Performance - Indoor Dry Bulb Temperature at 341 CFM																					
ODU DB(°F)	IDU WB (°F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB (°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-22°F (-30.0°C)	TC	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	
	S/T	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	
	PI	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	
-13°F (-25.0°C)	TC	9862	9857	9998	10039	10315	10289	10281	10274	10273	10265	10597	10583	10573	10572	10562	11177	11174	11151	11147	11124
	S/T	0.80	0.91	1.00	1.00	1.00	0.65	0.75	0.85	0.86	0.97	0.55	0.65	0.74	0.76	0.86	0.39	0.47	0.56	0.57	0.67
	PI	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60
-4°F (-20.0°C)	TC	10620	10615	10767	10811	11109	11081	11071	11064	11063	11054	11412	11397	11387	11385	11374	12037	12034	12009	12005	11979
	S/T	0.80	0.92	1.00	1.00	1.00	0.65	0.76	0.85	0.87	0.98	0.55	0.65	0.74	0.76	0.86	0.39	0.48	0.56	0.57	0.67
	PI	0.63	0.63	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.63	0.63	0.63	0.63	0.63
0°F (-17.8°C)	TC	10973	10966	11170	11216	11524	11443	11430	11422	11420	11535	11779	11763	11748	11746	11741	12413	12410	12409	12411	12419
	S/T	0.81	0.93	1.00	1.00	1.00	0.66	0.77	0.86	0.88	1.00	0.55	0.66	0.75	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.64	0.64	0.64	0.64	0.64
5°F (-15.0°C)	TC	11287	11326	11585	11638	11997	11817	11819	11820	11832	12009	12194	12197	12200	12200	12217	12910	12906	12912	12914	12920
	S/T	0.82	0.94	1.00	1.00	1.00	0.66	0.78	0.87	0.88	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.59	0.69
	PI	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.66	0.66	0.66	0.66	0.66
14°F (-10.0°C)	TC	11257	11252	11481	11535	11893	11816	11806	11798	11796	11907	12221	12206	12193	12191	12179	12995	12991	12966	12961	12934
	S/T	0.81	0.93	1.00	1.00	1.00	0.65	0.76	0.86	0.87	1.00	0.55	0.66	0.75	0.76	0.87	0.39	0.48	0.56	0.57	0.67
	PI	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69
17°F (-8.3°C)	TC	11221	11258	11518	11574	11950	11780	11785	11787	11797	11963	12182	12186	12191	12192	12207	12955	12952	12950	12952	12963
	S/T	0.82	0.93	1.00	1.00	1.00	0.66	0.77	0.87	0.88	1.00	0.55	0.66	0.76	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.71	0.71	0.71	0.71	0.71
23°F (-5.0°C)	TC	11509	11558	11857	11916	12309	12092	12095	12097	12113	12322	12509	12513	12516	12517	12537	13301	13297	13306	13308	13315
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.89	0.39	0.48	0.57	0.59	0.69
	PI	0.71	0.71	0.71	0.71	0.72	0.71	0.71	0.71	0.71	0.72	0.72	0.72	0.72	0.72	0.72	0.73	0.73	0.73	0.73	0.73
35°F (1.7°C)	TC	12197	12198	12269	12304	12701	12767	12767	12768	12768	12802	13167	13166	13167	13167	13169	13909	13905	13905	13904	13905
	S/T	0.78	0.88	0.97	0.99	1.00	0.63	0.73	0.82	0.83	0.93	0.54	0.63	0.72	0.73	0.83	0.39	0.46	0.54	0.56	0.65
	PI	0.69	0.69	0.69	0.69	0.72	0.73	0.73	0.73	0.73	0.73	0.76	0.76	0.76	0.76	0.76	0.82	0.82	0.82	0.82	0.82
41°F (5.0°C)	TC	12071	12071	12167	12252	12614	12633	12633	12633	12677	13027	13027	13026	13027	13027	13028	13758	13754	13753	13753	13754
	S/T	0.78	0.89	0.98	1.00	1.00	0.64	0.74	0.83	0.84	0.94	0.54	0.64	0.72	0.74	0.84	0.39	0.47	0.55	0.56	0.66
	PI	0.55	0.55	0.56	0.56	0.59	0.59	0.59	0.59	0.59	0.59	0.62	0.62	0.62	0.62	0.62	0.67	0.67	0.67	0.67	0.67
47°F (8.3°C)	TC	11945	11953	12113	12167	12527	12498	12498	12498	12556	12887	12885	12886	12886	12886	12886	13608	13605	13604	13603	13604
	S/T	0.79	0.90	1.00	1.00	1.00	0.64	0.75	0.84	0.85	0.96	0.54	0.65	0.73	0.75	0.85	0.39	0.47	0.55	0.57	0.67
	PI	0.42	0.42	0.43	0.43	0.45	0.45	0.45	0.45	0.45	0.47	0.47	0.47	0.47	0.47	0.47	0.52	0.52	0.52	0.52	0.52
50°F (10.0°C)	TC	11554	11872	12272	12343	12822	12206	12213	12297	12336	12834	12682	12699	12705	12706	12828	13556	13592	13627	13631	13648
	S/T	0.90	1.00	1.00	1.00	1.00	0.71	0.85	0.96	0.99	1.00	0.59	0.72	0.83	0.85	0.99	0.40	0.50	0.60	0.62	0.74
	PI	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48
59°F (15.0°C)	TC	11024	11411	11816	11888	12372	11664	11680	11813	11901	12384	12145	12158	12162	12179	12394	13042	13068	13094	13097	13108
	S/T	0.92	1.00	1.00	1.00	1.00	0.73	0.86	1.00	1.00	1.00	0.60	0.73	0.85	0.86	1.00	0.40	0.51	0.61	0.63	0.76
	PI	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.52	0.52	0.51	0.51	0.51
68°F (20.0°C)	TC	10493	10941	11347	11419	11903	11108	11134	11359	11431	11915	11589	11597	11618	11628	11925	12487	12510	12530	12532	12538
	S/T	0.95	1.00	1.00	1.00	1.00	0.74	0.89	1.00	1.00	1.00	0.61	0.75	0.87	0.89	1.00	0.40	0.51	0.63	0.65	0.78
	PI	0.53	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.55	0.55	0.55	0.55	0.55
77°F (25.0°C)	TC	9957	10437	10845	10918	11405	10556	10590	10857	10930	11417	11034	11047	11075	11087	11426	11917	11954	11981	11984	11995
	S/T	0.97	1.00	1.00	1.00	1.00	0.75	0.90	1.00	1.00	1.00	0.61	0.76	0.88	0.90	1.00	0.40	0.52	0.63	0.65	0.79
	PI	0.56	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.58	0.58	0.58	0.58	0.58
82°F (27.8°C)	TC	9836	10339	10756	10829	11324	10424	10464	10765	10841	11336	10908	10922	10956	10970	11346	11794	11839	11870	11873	11886
	S/T	0.99	1.00	1.00	1.00	1.00	0.76	0.91	1.00	1.00	1.00	0.62	0.77	0.89	0.92	1.00	0.40	0.52	0.64	0.66	0.80
	PI	0.63	0.63	0.63	0.63	0.64	0.63	0.63	0.63	0.63	0.64	0.63	0.63	0.63	0.63	0.64	0.64	0.64	0.64	0.64	0.64
86°F (30.0°C)	TC	9995	10493	10923	10999	11511	10540	10589	10935	11011	11523	11040	11052	11093	11109	11533	11952	11998	12026	12029	12040
	S/T	1.00	1.00	1.00	1.00	1.00	0.77	0.93	1.00	1.00	1.00	0.62	0.78	0.91	0.93	1.00	0.40	0.52	0.64	0.66	0.81
	PI	0.73	0.74	0.74	0.74	0.75	0.74	0.74	0.74	0.74	0.75	0.74	0.74	0.74	0.74	0.75	0.75	0.75	0.75	0.75	0.75
95°F (35.0°C)	TC	10277	10332	10666	10729	11153	10922	10928	10932	10945	11167	11391	11393	11400	11401	11420	12274	12269	12272	12275	12292
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.88	0.				

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Heating Performance - Indoor Dry Bulb Temperature at 341 CFM										
Outdoor DB(°F)	TC Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-22°F(-30.0°C)	/	/	/	/	/	/	/	/	/	/
-13°F(-25.0°C)	5128	5025	5000	4980	4935	0.92	0.93	0.93	0.93	0.93
-4°F(-20.0°C)	6256	6131	6100	6076	6021	0.96	0.97	0.97	0.97	0.98
0°F(-17.8°C)	7122	6979	6944	6917	6854	1.01	1.02	1.02	1.02	1.02
5°F(-15.0°C)	8204	8040	8000	7968	7896	1.06	1.07	1.08	1.08	1.08
14°F(-10.0°C)	9666	9472	9425	9387	9303	1.10	1.11	1.11	1.11	1.12
17°F(-8.3°C)	10153	9950	9900	9860	9772	1.11	1.12	1.12	1.13	1.13
22°F(-5.6°C)	10306	10100	10049	10009	9919	1.13	1.13	1.14	1.14	1.14
27°F(-2.8°C)	10459	10250	10199	10158	10066	1.14	1.15	1.15	1.15	1.16
32°F(0.0°C)	10612	10400	10348	10306	10214	1.15	1.16	1.16	1.16	1.17
35°F(1.7°C)	10704	10490	10437	10396	10302	1.16	1.17	1.17	1.17	1.17
37°F(2.8°C)	11364	11137	11081	11037	10937	1.18	1.19	1.19	1.19	1.20
42°F(5.6°C)	13015	12754	12691	12640	12526	1.24	1.25	1.25	1.25	1.26
44.6°F(7.0°C)	13675	13401	13334	13281	13161	1.26	1.27	1.27	1.28	1.28
47°F(8.3°C)	14665	14372	14300	14243	14115	1.30	1.31	1.31	1.31	1.32
52°F(11.1°C)	14509	14219	14148	14091	13964	1.15	1.16	1.16	1.16	1.17
57°F(13.9°C)	14352	14065	13995	13939	13814	1.00	1.01	1.01	1.02	1.02
59°F(15.0°C)	14290	14004	13934	13878	13753	0.95	0.95	0.96	0.96	0.96
62°F(16.7°C)	14196	13912	13843	13787	13663	0.86	0.87	0.87	0.87	0.87
64.4°F(18.0°C)	14121	13839	13769	13714	13591	0.79	0.79	0.80	0.80	0.80
69.8°F(21.0°C)	13952	13673	13605	13550	13428	0.63	0.64	0.64	0.64	0.64
75.2°F(24.0°C)	13783	13508	13440	13386	13266	0.47	0.48	0.48	0.48	0.48

Bolded values were tested. All other values are simulated and for reference only.